

Docker Containerization Guide IBM Telco Customer Churn Prediction

Machine Learning Operations

February 9, 2026

Contents

1 Introduction to Docker

Docker is a containerization platform that packages an application with all its dependencies into a single standardized unit called a container.

1.1 Why Docker?

- Reproducibility: Same environment everywhere
- Isolation: No conflicts with system packages
- Portability: Run on any machine with Docker
- Scalability: Easy deployment
- Simplicity: One command to run

2 Project Structure

```
telco-churn-xgboost-shap/  
  Dockerfile  
  api.R  
  TRAIN_IBM_TELCO_85.R  
  xgboost_churn_model.rds  
  WA_Fn-UseC_-Telco-Customer-Churn.csv
```

3 Understanding the Dockerfile

3.1 Base Image

```
FROM rocker/r-ver:4.3.2
```

Start from R 4.3.2 pre-installed image.

3.2 Install System Dependencies

```
RUN apt-get update && apt-get install -y \  
  libcurl4-openssl-dev \  
  libssl-dev \  
  libxml2-dev \  
  libsodium-dev
```

Install libraries required by R packages.

3.3 Install R Packages

```
RUN R -e "install.packages(c('plumber', 'xgboost'), \  
  repos='http://cran.us.r-project.org')"
```

Install Plumber (API framework) and XGBoost (ML library).

3.4 Setup Working Directory

```
WORKDIR /app
COPY api.R .
COPY xgboost_churn_model.rds .
```

Copy application files into container.

3.5 Expose Port and Start Service

```
EXPOSE 8000
CMD ["R", "-e", "plumber::plumb('api.R')\n$run(port=8000)"]
```

Listen on port 8000 and start the API server.

4 Building the Docker Image

4.1 Build Command

```
docker build -t telco-churn:latest .
```

Parameters:

- `docker build`: Create new image
- `-t telco-churn:latest`: Tag image
- `.`: Use Dockerfile from current directory

5 Running the Container

5.1 Run Command

```
docker run -p 8000:8000 \\  
  --name telco-api \\  
  telco-churn:latest
```

Parameters:

- `-p 8000:8000`: Map port 8000
- `--name`: Container name
- `telco-churn:latest`: Image to run

6 Docker Commands Reference

6.1 Image Management

```
docker images          # List images
docker rmi telco-churn:latest # Remove image
docker inspect telco-churn # View details
```

6.2 Container Management

```
docker ps          # List running containers
docker ps -a       # List all containers
docker stop telco-api # Stop container
docker logs telco-api # View logs
docker rm telco-api # Remove container
```

7 API Usage

7.1 Health Check

```
curl http://localhost:8000/health
```

7.2 Make Prediction

```
curl -X POST http://localhost:8000/predict \\\
  -H "Content-Type: application/json" \\\
  -d '{"gender": 0, "tenure": 24, ...}'
```

8 Docker vs Direct Execution

Aspect	Direct	Docker
Setup	Quick	Slow first build
Dependencies	Possible issues	None
Reproducibility	Difficult	Guaranteed
Deployment	Manual	Automated
Portability	Limited	Excellent

9 Quick Start

9.1 Without Docker

```
cd /home/prajwa/-telco-churn-xgboost-shap-
Rscript -e "plumber::plumb('api.R')\$run(port=8000)"
```

9.2 With Docker

```
docker build -t telco-churn .  
docker run -p 8000:8000 telco-churn
```